7.8 Answers: graphs and averages

← Unit 7.8 · Practice

For the “talk about it” and critical-thinking questions, any well-reasoned answer counts. The notes give the main points to listen for.

Set 1 · Warm-up: bar chart

1. 14
2. 6
3. Bike
4. 8 (14 − 6)
5. 26 (14 + 6 + 4 + 2)
6. 6 (4 + 2)
7. 12 (26 − 14)
8. Bus

Talk about it: lots of cycling and walking suggests the school is close to home, which could be a town or a village. You can’t be sure from one graph.

Set 2 · Building: line graph

1. 15:00 (18 °C)
2. 8 degrees (16 − 8)
3. 7 degrees (18 − 11)
4. Between 09:00 and 12:00 (5 degrees; the line is steepest there)
5. 10 degrees (18 − 8)
6. 13 °C (8 + 11 + 16 + 18 + 14 + 11 = 78, and 78 : 6 = 13)
7. 09:00 and 21:00 (both 11 °C)
8. 16 °C

Talk about it: a line graph shows how something changes over time. A pie chart shows parts of one whole, and temperatures aren’t parts of anything.

Set 3 · Building: average

1. 5 (10 : 2)
2. 5 (15 : 3)
3. 25 (100 : 4)
4. 6 (30 : 5)
5. € 6 (€ 24 : 4)
6. 4 km (20 : 5)
7. 7 (35 : 5)
8. 1,8 m (5,4 : 3; or notice that 1,5 and 2,1 are equally far from 1,8)
9. 11 (the four numbers must add up to 4 × 10 = 40; 8 + 12 + 9 = 29; 40 − 29 = 11)
10. 5

Talk about it: the average is what each day *would* be if the kilometres were shared out fairly. It doesn’t have to be one of the real numbers.

Set 4 · Building: pie chart

1. 20 (half of 40)
2. 10 (a quarter of 40)
3. 5 (an eighth of 40)
4. 25%

5. 40 (half of 80)
6. 1/4 (10/40, divide top and bottom by 10)
7. 10 children (5 + 5), which is a quarter of the circle (two eighths)
8. Apple

Set 5 · Stretch: thinking critically

1. 4 cones
2. About 5 times as tall (the vanilla bar shows 1 step above 95, the chocolate bar 5 steps)
3. No. 100 is only a little more than 96. The vertical axis starts at 95 instead of 0, which makes a small difference look huge.
4. 90%
5. Only 10 children were asked, which is very few; and they were asked at a CrunchyChips party, so they were probably fans already (and maybe didn't want to be rude). Other good answers: we don't know what they were compared with, or what the question was.
6. 15 years (10 + 10 + 10 + 10 + 10 + 40 = 90; 90 : 6 = 15)
7. Nobody in the group is 15. One very different number (the teacher's age) pulls the average up a lot. An average can hide what the group is really like.
8. About 4 times as big (a 2 × 2 square is made of 4 small squares), even though sales only doubled.
9. 100

Puzzle corner

- Five numbers with average 6 and no 6: the total must be 30, for example 5, 5, 5, 7, 8 or 1, 2, 3, 4, 20.
- All different and below 10: for example 2, 4, 7, 8, 9 (total 30).
- No. The average is "levelling out", so it is always between the smallest and the largest number (or equal to them if all the numbers are the same).
- You need a 9. To average 7,5 over four tests the total must be 30; after three tests averaging 7 the total is 21; 30 − 21 = 9.